Fifteen-week window for recurrent muscle strains in football: a prospective cohort of 3600 muscle strains over 23 years in professional Australian rules football

John W Orchard ,[1] Mohammad Chaker Jomaa ,[1] Jessica J Orchard ,[1] Katherine Rae,[2] Daniel Tyler Hoffman ,[3] Tom Reddin ,[4] Tim Driscoll [1]

[1]School of Public Health, University of Sydney Medical School, Sydney, New South Wales, Australia
[2]The Sports Clinic, University of Sydney, New South Wales, Australia
[3]School of Exercise and Nutrition Sciences, Centre for Sport Research, Deakin University Faculty of Health, Burwood, Victoria, Australia
[4]Sydney University Australian National Football Club, Camperdown, New South Wales, Australia

Correspondence to
Dr John W Orchard, School of Public Health, University of Sydney Sydney Medical School, Sydney, NSW 2006, Australia; john.orchard@sydney.edu.au

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ABSTRACT

Objectives To determine the rates of muscle strain injury recurrence over time after return to play in Australian football and to quantify risk factors.

Methods We analysed Australian Football League player data from 1992 to 2014 for rates of the four major muscle strain injury types (hamstring, quadriceps, calf and groin) diagnosed by team health professionals. Covariates for analysis were: recent history (≤8 weeks) of each of the four muscle strains; non-recent history (>8 weeks) of each; history of hip, knee anterior cruciate ligament, knee cartilage, ankle sprain, concussion or lumbar injury; age; indigenous race; match level and whether a substitute rule was in place.

Results 3647 (1932 hamstring, 418 quadriceps, 458 calf and 839 groin) muscle strain injuries occurred in 272 759 player matches. For all muscle strains combined, the risk of injury recurrence gradually reduced, with recurrence risks of 9% (hamstring), 5% (quadriceps), 2% (calf) and 6% (groin) in the first match back and remaining elevated for 15 weeks after return to play. The strongest risk factor for each muscle injury type was a recent history of the same injury (hamstring: adjusted OR 13.1, 95% CI 11.5 to 14.9; calf OR 13.3, 95% CI 9.6 to 18.4; quadriceps: OR 25.2, 95% CI 18.8 to 33.8; groin OR 20.6, 95% CI 17.0 to 25.0), followed by non-recent history of the same injury (hamstring: adjusted OR 3.5, 95% CI 3.2 to 3.9; calf OR 4.4, 95% CI 3.6 to 5.4; quadriceps OR 5.2, 95% CI 4.2 to 6.4; groin OR 3.5, 95% CI 3.0 to 4.0). Age was an independent risk factor for calf muscle strains (adjusted OR 1.6, 95% CI 1.3 to 2.0). Recent hamstring injury increased the risk of subsequent quadriceps (adjusted OR 1.8, 95% CI 1.2 to 2.7) and calf strains (OR 1.8, 95% CI 1.2 to 2.6). During the 'substitute rule' era (2011–2014), hamstring (adjusted OR 0.76, 95% CI 0.67 to 0.86), groin (OR 0.78, 95% CI 0.65 to 0.93) and quadriceps (OR 0.70, 95% CI 0.53 to 0.92) strains were less likely than outside of that era but calf (OR 1.6, 95% CI 1.3 to 1.9) strains were more likely than before the substitute rule era.

Conclusion Recent injury is the greatest risk factor for the four major muscle strains, with increased risk persisting for 15 weeks after return to play.

INTRODUCTION

Muscle strains are the most common injuries in football sports. The hamstring strain is the most common injury in elite football (soccer),[1] Australian rules football[2] and also a very common injury in American football,[3] rugby league[4] and rugby union.[5] Quadriceps strains,[6] calf strains[7] and groin injuries[8] are common in all codes of football (including Australian football), and groin injuries are also particularly common in ice hockey.[9]

The most common risk factor for muscle injury is the history of injury to the same muscle group.[2 7 10] Other risk factors for the various muscle strains include: muscle weakness,[10] previous anterior cruciate ligament (ACL) injury,[10] fixture congestion,[11] increase in recent high speed running[12] and indigenous race.[10] The two study cohorts with the highest sample sizes for analysis of muscle strain risk factors, the Australian Football League (AFL) injury database[2] and the Union of European Football Associations injury database,[1 13 14] share remarkable similarity in the distribution of injuries and risk factors.[2 15]

We aimed to use the AFL injury database to further assess risk factors for muscle strain injury—hamstring, thigh, groin and calf muscle—in the professional AFL. We had a specific goal to scrutinise the relationship between index injury and recurrence risk over time when the player returned to competition.

METHODS

Data were extracted from the AFL injury database from 1992 to 2014. The methods are similar to those used in an earlier study from the same injury surveillance system involving a smaller cohort.[2]

An injury was defined as a condition which was severe enough to cause a player to miss a match in either the regular season or finals in accordance with the injury definition use by the AFL Injury Surveillance System. For this analysis, groin injuries (which are predominantly muscle strains) were added to hamstring, quadriceps and calf strains. A muscle strain was defined as an injury to a clinically identifiable muscle group, excluding direct contact injuries such as lacerations and contusions.

We have used the traditional term 'muscle strain' (rather than 'muscle injury') to make it clear that we are including intrinsic injuries but not muscle injuries involving direct contact.

Logistic regression analysis

The unit for analysis in logistic regression was the 'player match'. All player matches (AFL or local league) from 1992 to 2014 inclusive were analysed, except the scheduled final round of each player's season. The final round was excluded as players were far less likely to suffer an injury that caused them to miss a subsequent match.

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Binary logistic regression (multivariable analysis) was used, incorporating a forward stepwise method, with a $p<0.05$ required to enter the equation and a $p>0.10$ required to remove a variable from the equation, using IBM SPSS V.24. The dependent variables selected were the occurrence of any of the four muscle strain injury categories in a match.

Covariates for analysis, all either binary or converted to binary, were:

1. Recent history of either hamstring, quadriceps, calf or groin muscle strains (four variables) within the previous 8 weeks.[2]
2. Non-recent history of any of the four muscle strains (four variables) occurring longer than 8 weeks ago.
3. Histories of either hip injury, knee ACL injury, knee cartilage injury, ankle sprain, concussion or lumbar injury (six variables) having caused a missed game while on an AFL list.
4. Age (stratified into 24 years and over or 23 years and under to become a binary variable).
5. Indigenous race.
6. Match level: AFL or local league.
7. Substitute rule in place (relevant for seasons 2011–2014 in this dataset), where the four man interchange bench was divided into three interchange players and one substitute player.

Analysis of recurrence rates over time

Based on previous studies on muscle strains, it was expected that recent and non-recent history of the same muscle strain would be the main risk factors in the equation. These two risk factors had been differentiated with a cut-off of 8 weeks from the initial injury (≤8 weeks being considered a recent injury and >8 weeks being considered a past or non-recent injury). Eight weeks was chosen in keeping with consensus statements from other football codes that injury recurrences within 2 months are considered 'early recurrences'.[16 17]

Week-by-week analysis of recurrence rates was performed to examine the risk of reinjury over time. This was done by plotting 'weeks since return from injury' (with week one being the first return match) and percentage of players who suffered an injury recurrence to the same muscle group (using the total number of players at the same week of return, both those injured and those who survived the game, as the denominator). An analysis was undertaken of the relationship between recurrence rates over time, with linear and logarithmic trendlines considered, aiming for a trendline of best fit (Microsoft Excel, Seattle USA). Each type of muscle strain was considered at elevated risk of recurrence until the week when the logarithmic trendline reached the baseline risk for a player with a history of that injury in a prior season.

Patient and public involvement

This study did not include patient and public involvement.

RESULTS

There were 3647 (1932 hamstring, 418 quadriceps, 458 calf and 839 groin) muscle strain injuries, both new and recurrent, occurring in 272 759 player matches, played by 3200 players, from 1992 to 2014.

For all injuries, the strongest risk factor was a recent history of the same injury, followed by non-recent history of the same injury (table 1). Age was an independent risk factor for calf strains (even when adjusted for injury history), but not for other

Table 1 Significant muscle strain match injury risks after binary logistic regression analysis

Muscle strain type	Risk factor*	Unadjusted OR (single variable)	Adjusted OR (multivariable)*	95% CIs (for adj OR)		P value
Hamstring	Recent hamstring	7.96	13.10	11.52	14.89	<0.001
	Non-recent history hamstring	2.34	3.49	3.15	3.86	<0.001
	Recent calf	1.63	1.49	1.04	2.11	0.03
	Indigenous player	1.65	1.34	1.16	1.55	<0.001
	Past ACL	1.55	1.23	1.03	1.46	0.02
	Past concussion	0.91	0.77	0.63	0.94	0.01
	Substitute rule (2011–2014)	0.78	0.76	0.67	0.86	<0.001
Calf	Recent calf	10.66	13.30	9.62	18.38	<0.001
	Non-recent history of calf	4.54	4.38	3.56	5.39	<0.001
	Recent hamstring	1.89	1.75	1.17	2.60	0.01
	Age 24 or more	2.27	1.62	1.32	1.98	<0.001
	Substitute rule	1.89	1.59	1.30	1.94	<0.001
	Past knee cartilage	0.94	0.68	0.48	0.96	0.03
Quadriceps	Recent quad	17.70	25.19	18.75	33.84	<0.001
	Past not recent quadriceps	3.86	5.16	4.17	6.38	<0.001
	Recent hamstring	2.00	1.80	1.20	2.71	0.01
	Local league match	1.36	1.50	1.23	1.83	<0.001
	Substitute rule	0.68	0.70	0.53	0.92	0.01
Groin	Recent groin	14.72	20.62	17.02	24.99	<0.001
	Past not recent groin	2.37	3.46	2.97	4.03	<0.001
	Local league match	1.35	1.36	1.19	1.57	<0.001
	Past ankle	0.90	0.82	0.69	0.98	0.03
	Substitute rule	0.80	0.78	0.65	0.93	0.01

*All risk factors being binary are compared with the opposite condition as the reference group (ie, recent hamstring strain OR is the odds of injury for players who have suffered a recent hamstring strain compared with all players who have not suffered a recent hamstring strain).
ACL, anterior cruciate ligament.

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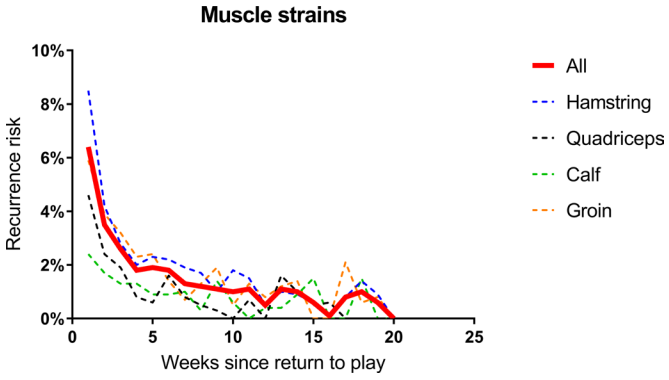

Figure 1 Muscle strain recurrence risk over time.

muscle injuries (table 1). Indigenous race was associated with hamstring strains, but not the other types of muscle strain. In the substitute rule era (2011–2014), hamstring, quadriceps and groin strains were less likely but calf strains were more likely. Past ACL injury was associated with higher risk of hamstring strain. Some of the relationships seen between non-muscle strain injuries and muscle strains appeared to be protective (past knee cartilage injury associated with lower risk of calf strain; past concussion associated with lower risk of hamstring strain; past ankle injury associated with lower risk of groin strain).

Relationships between muscle strain injury risk of recurrence over time, in weeks since return to play, are shown in figures 1–5. Figure 1 plots all four muscle strains collectively, showing a very similar pattern for the individual curves, while figures 2–5 plot individual muscle strain recurrence rates over time. These Figures show that all the muscle strain types had their highest risk of recurrence in the initial match on return from injury. For example, the absolute risk of recurrence in the first week was 9% for hamstring strains, 6% for groin strains, 5% for quadriceps strains and 2% for calf strains. All four muscles strain types had the second highest risk of recurrence in the second match back from injury (hamstring 4%, groin 4%, quadriceps and calf 2%).

The week-by-week recurrence rates tended to slowly drop each week. All four muscle strains exhibited a log decay in recurrence risk over time with moderate-to-high correlation of best fit (R^2 for hamstring 0.81, calf 0.52, quadriceps 0.71, groin 0.79 and all muscle strains collectively 0.85). For all four muscle strains, the best fit trend line was at higher risk than baseline (for a player with a past injury history) for many weeks. Specifically, hamstring strains remained at elevated risk for 15 weeks, quadriceps for 15 weeks, calf strains for 18 weeks and groin strains for 19 weeks. Hamstring injuries have a higher early risk of recurrence than the other three groups. Calf injuries have the least correlation with the log decay curve, showing the lowest risk of early recurrence of the four groups, but with more late recurrences.

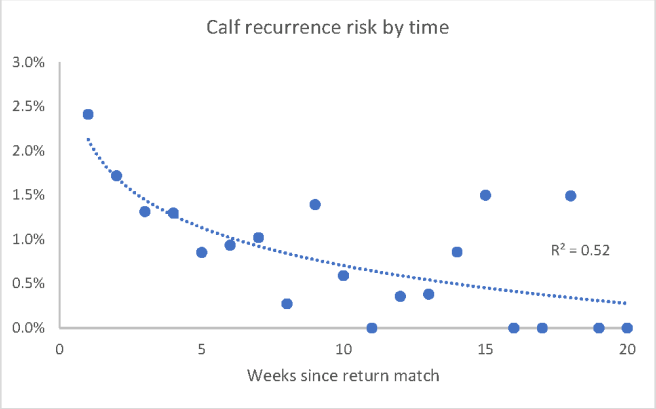

Figure 3 Calf recurrence risk over time.

DISCUSSION

This is the largest analytical study of sport-associated muscle strain risk factors published to date. It includes data for 3647 muscle strains occurring in 272 759 player matches, superseding the size of previous studies.[15 18]

This study expands on previous work[19] to demonstrate that there is an extended period of increased risk for recurrence of muscle strains, persisting for at least 15 weeks after return to play. It confirms that recent and non-recent history of injury of the same muscle group are by far the strongest risk factors for muscle strains, consistent with previous findings.[2 7 10 15] In addition, it quantifies recurrence rates, which were highest in the first match back: 9% (hamstring), 5% (quadriceps), 2% (calf) and 6% (groin).

High recurrence risk for players with an injury history

The ORs for recent (within the past 8 weeks) and past muscle strains are the highest published to date. Adjusted ORs for recent history of the same injury were: hamstring OR 13.10; calf OR 13.30; quadriceps: OR 25.19; groin OR 20.62. Adjusted ORs for past (non-recent) history of the same injury were hamstring OR 3.49; calf OR 4.38; quadriceps OR 5.16; groin OR 3.46. These ratios are higher than the previous study using a smaller AFL cohort.[2]

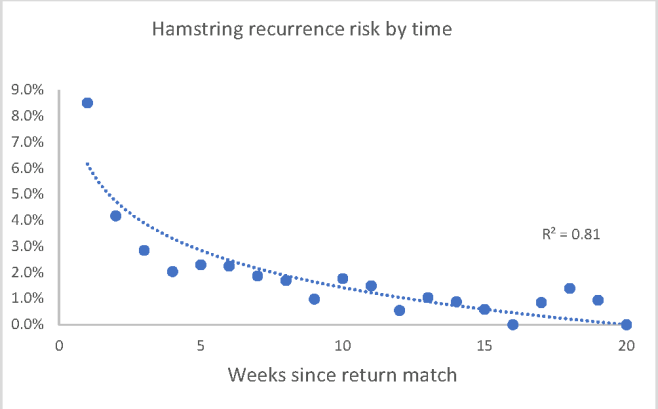

Figure 2 Hamstring recurrence risk over time.

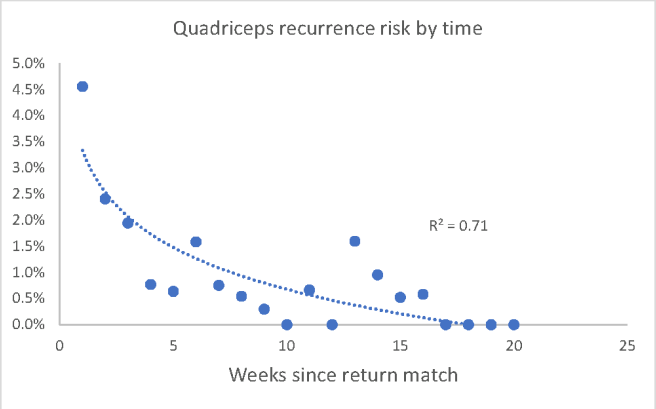

Figure 4 Quadriceps recurrence risk over time.

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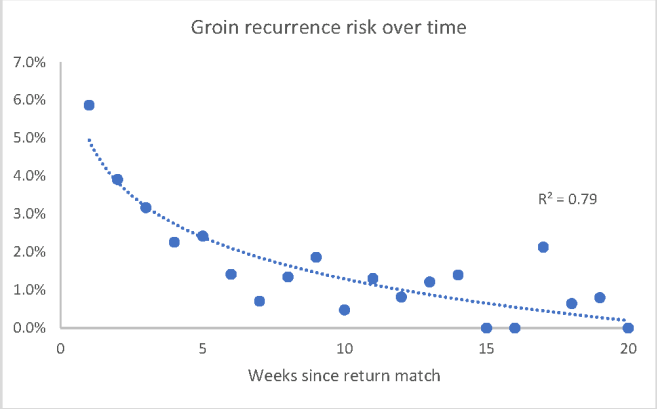

Figure 5 Groin recurrence risk over time.

To give perspective to the ORs, the absolute risk of sustaining a hamstring strain in a single game (single variable analysis, not adjusted for confounding) was approximately 0.2% for a player with no history, 0.7% for a player with a history in a prior season, 1.4% for a player with a history prior to the previous 8 weeks, and 4.0% for a player with a recent history (in the past 8 weeks).

The high ORs are likely to be due to the very low absolute risks for players who had not previously suffered a muscle strain of that type. These absolute risks were 0.22% risk per player per game for hamstring strain, 0.07% for calf, 0.06% for quadriceps and 0.12% for groin. The likely explanation for the low absolute risks is that history was well captured for this study as it spanned over 20 years. The AFL is a competition with no rival leagues and most players in the injury database would have had their entire adult career in the league captured by this study (other than a very small number of players in the early 1990s whose careers started in the 1980s). Therefore, it is likely that this study correctly identified the relatively low muscle strain rates of a player without a history of that type of muscle strain.

Interpreting risk factors

It is difficult to interpret long-term trends in muscle strain incidences as there are many confounders,[18] although it is pleasing that in recent seasons the risks of most of the muscle strain injury types were decreasing in the AFL.

Some significant risk factors are likely to represent true relationships, whereas others may represent residual confounding. Recent hamstring injury increased the risk of subsequent quadriceps and calf strains. This may be as a direct result of a reduction in stride length.[20]

Residual confounding is more likely to explain concussion history appearing to 'protect' against hamstring strain. We consider this is more likely to be a marker of playing style (eg, 'inside' players more at risk of concussion vs 'outside' players more at risk of hamstring strain). While there is recent evidence that a history of concussion increases susceptibility for other injuries,[21] we did not find this for muscle strains in AFL players (and in fact found the opposite for hamstring strains).

We are unsure of specific mechanisms of how past knee cartilage injury would protect against calf strains or ankle injury would protect against groin strains. It is possible that a history of a knee cartilage injury may make posterior knee pain more likely to be diagnosed as a knee injury recurrence than an upper calf strain. In a similar fashion to hamstring/concussion, it is possible that a player with a playing style of avoiding contact may be more predisposed to groin injuries but less predisposed to ankle injuries.

In the substitute rule era, the risk of hamstring, quadriceps and groin strains occurring during matches all decreased but the risk of calf strains significantly increased. While the substitute rule may have been partially responsible for these effects, additional confounding cannot be ruled out. An earlier study of AFL hamstring strains prior to the introduction of the substitute rule showed that these injuries had increased in the 2000s along with increasing team interchange rates.[22] In an attempt to limit the ongoing increase in player interchanges, the AFL introduced the substitute rule in 2011,[23] meaning that only three of the bench players were allowed to be interchanged and the fourth player needed to be a permanent substitute. The AFL had previously increased the number of players on the interchange bench from three to four in 1998.[23] The substitute rule was removed at the end of 2014 and replaced by limits on number of interchanges that each team can make in a match.

A number of similar risk factors were identified both in this study and the other large cohort study of muscle strains in a different football code.[24] These include: high risk of recurrences, that injuries to one muscle group increased risk of injuries to another muscle group and that player age was independently a risk factor only for calf strains.[24] A recent study from Australian football also found that risk of subsequent injury (different injury to index injury) was elevated for many weeks after return to play.[25]

Methodological considerations

The major strength of the study was the large database, giving high power and associated narrow confidence intervals. Limitations include the injury and category definitions, that the side of injury was not analysed, that the history for all injuries was highly correlated with player age, and some definitional uncertainty of diagnoses (eg, crossover between groin muscle strain and hip injuries). Another factor not analysed by this study was fixture congestion, as AFL teams generally play one match (only) per week. There were also other potential risk factors for muscle strain injuries that were not included in this study, including an unaccounted history of injury prior to becoming an AFL listed player (and therefore not captured by the database), player workload, muscle weakness[26] and speed of running.

Side of injury and specific muscle within each group were not analysed because of an incomplete data set. For example with some 'recurrences', it was not noted in either the initial or subsequent muscle strain on which side the injury recurred. It could also be argued that it would be invalid to analyse left hamstring and right hamstring for a player as if they were independent muscles, and similarly whether, say, rectus femoris and vastus muscles are independent muscles on the same side.

The interaction between history and age is difficult to account for. It is noteworthy that this study found that both age and history were independent risk factors for calf strains, consistent with previous research.[24] This study also found (consistent with previous research) that once history was taken into account, age was no longer a significant risk factor for hamstring strain.[24] Of all the variables considered in this analysis, age was the variable which was perhaps oversimplified by conversion to binary. However, given its lack of prominence in the results it was not necessary to further analyse it as a continuous variable for hamstring, quadriceps and groin strains.

Age was a significant risk factor for calf strain, so further analysis could consider age as a continuous variable alongside history (which is related and also highly relevant). It is also noteworthy that calf strains are also the only individual muscle strain group with only a moderate correlation with the log decay line of best

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fit (all other muscle strains had a very high correlation of injury recurrence risk with the number of weeks since return to play). Calf strains also increased in risk in the last 4 years of the study (2011–2014) when the substitute rule was in place in the AFL, whereas the other muscle strains decreased in risk. It could be speculated that there may be a factor which relates to age that results in calf strains behaving differently to the other muscle strains.

There are also potential effects arising from inclusion of each player multiple times within the analysis. This occurred because players were included for each match they played. Since the 'career risk' of suffering a muscle strain was so high, we considered that the 'player-week' risk (while introducing dependency between some observations) was the most appropriate unit. The effect of this dependency may have been to make the confidence intervals narrower than they might have been if the dependency was formally taken into account. However, it should not have affected the overall key results of the study.

A limitation of the analysis is the risk of sparse data bias.[27] The dataset overall is very large with high unadjusted ORs for recent injuries, indicating that there is a substantial increase in likelihood of injury in the setting of a recent injury of the same type. In theory, adjusting for confounders gives a more accurate impression of the true OR. The inclusion of many variables (eight significant binary variables with respect to hamstring strains) means that there are 256 (2^8) different outcomes from the eight included hamstring binary risk factors. Each binary combination will only average 7.5 hamstring strains (1932/256), meaning there is a possibility of sparse date bias.[27] This limitation is worth acknowledging but does not change the headline finding of very large increase in risk in the setting of recent muscle strain injury.

Despite these limitations, the very high correlations of muscle strain recurrence risk with the logarithmic decay curve of best fit strongly suggests that this is a fairly 'clean' database without substantial issues caused by uncertainty over definitions.

Although not available for this analysis, it is likely that real-time measurement of player movement would add significantly to the prediction model of injury risk for individual players. That is, that players who do more high speed runs, accelerations and, in particular, suddenly increase their amount of high speed running in a given match or week probably have higher risk of muscle strain injury, over and above the risk factors described in this study.[12] The real-time monitoring of player movement became commonplace among AFL teams in the latter years of the study, and this itself is a potential confounder for the observed lower rates of some muscle injuries in the period 2011–2014, as player movement monitoring itself may encourage moderation of loading and therefore be protective against some of the muscle strains.

A 'glass-half refilled' approach to the data presented could be used to argue that all four muscle strain types have less than 3% risk (per week) of recurrence by the fourth week (match) back from injury. We have actually highlighted the opposite in the abstract and title—that for at least 15 weeks the log decay plots still exhibited increased risk of reinjury compared with baseline (risk if a past history in a prior season). Our interpretation of the data are that for 15 weeks beyond return to play, the player in question is still 'carrying' the injury concerned, having a higher risk than baseline of reinjury.

CONCLUSION

This study confirms the primacy of recent history in particular as a risk factor far and above all other factors for muscle strains in footballers. It further establishes the paradigm that many footballers return to play while 'carrying' muscle strains that have not completely resolved and hence they remain at elevated risk of recurrence after apparently successful return to play. To this end, this study demonstrates that this risk of recurrence gradually reduces for all muscle injuries with recurrence risks of 9% (hamstring), 5% (quadriceps), 2% (calf) and 6% (groin) in the first match back, gradually reducing, but with elevated risk of injury recurrence persisting for 3 months after return to play. Careful management of players is warranted in the weeks after return to play given the persisting injury risk.

What are the findings?

- The muscle strain recurrence rates in this elite cohort are 9% (hamstring), 5% (quadriceps), 2% (calf) and 6% (groin) in the first match back. The increased risk of injury recurrence gradually reduces for 15 weeks after the injury.

How might it impact on clinical practice in the future?

- Careful management of players is warranted in the weeks after apparently successful return to play, given that injury risk is raised for an extended period.
- These data point to a 15-week window where increased risk of muscle strain persists. This is more than half a season in Australian Football. This fact should be shared with relevant parties when return to play decisions are made.

Twitter John W Orchard @DrJohnOrchard, Jessica J Orchard @jessicajorchard and Daniel Tyler Hoffman @D_Hoffman93

Contributors JWO is the first author of this paper. All subsequent authors have made substantial contributions to data analysis and interpretation as well as drafting and critical revision of the work.

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Competing interests JWO was the AFL injury surveillance coordinator between 1992 and 2014. KR was the GWS Giants AFL team medical officer in the 2018 season.

Patient consent for publication Not required.

Ethics approval The survey methods were approved by the AFL, the AFL Research Board, AFL Players Association and AFL Doctors Association, acting as Institutional Review for the study. The analysis only involved de-identified player data and was of very large groups of injuries (hence with no potential for an individual player to have been re-identified from data analysis), and therefore, did not require full Human Ethics assessment according to Australia's National Health and Medical Research Council guidelines and the ethics policy of the authors' institution (University of Sydney, Australia).

Provenance and peer review Not commissioned; externally peer reviewed.

Data availability statement Data may be obtained from a third party and are not publicly available. Raw data availability would require specific request to the AFL Research Board. Analysed data outputs, (Excel and SPSS outputs) but not raw data, would be available on reasonable request to the authors.

ORCID iDs
John W Orchard http://orcid.org/0000-0003-3530-1711
Mohammad Chaker Jomaa http://orcid.org/0000-0001-7916-2755
Jessica J Orchard http://orcid.org/0000-0002-5702-7277
Daniel Tyler Hoffman http://orcid.org/0000-0002-5147-5829
Tom Reddin http://orcid.org/0000-0002-8210-7623
Tim Driscoll http://orcid.org/0000-0003-0057-2490

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